

Demographic Inference from NGS data

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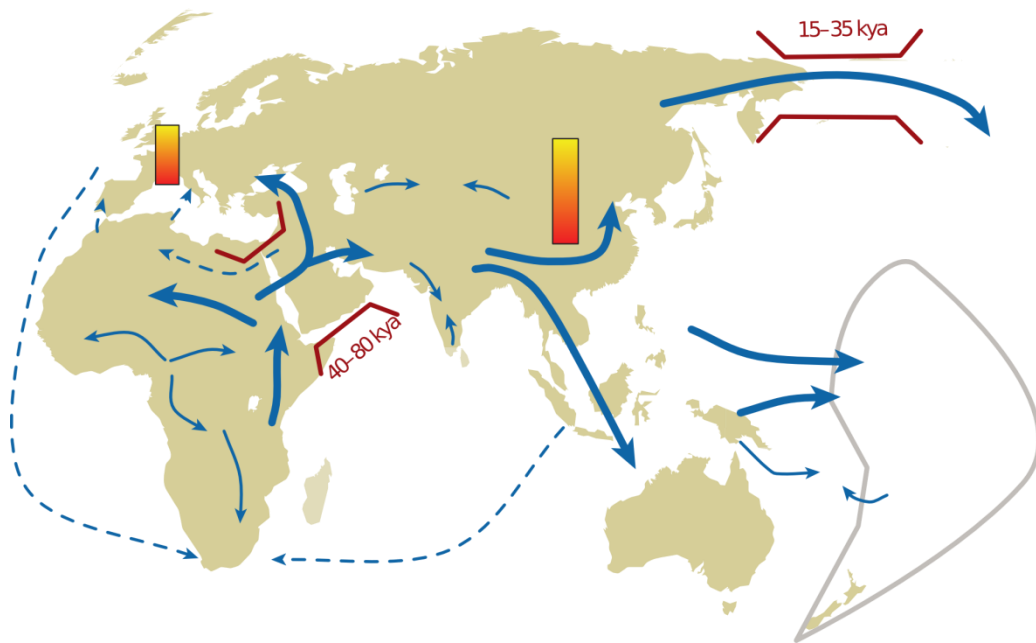
Introduction

- The current genetic diversity is the outcome of past evolutionary processes.
- Hence, we can use genetic diversity to tell stories about the past.

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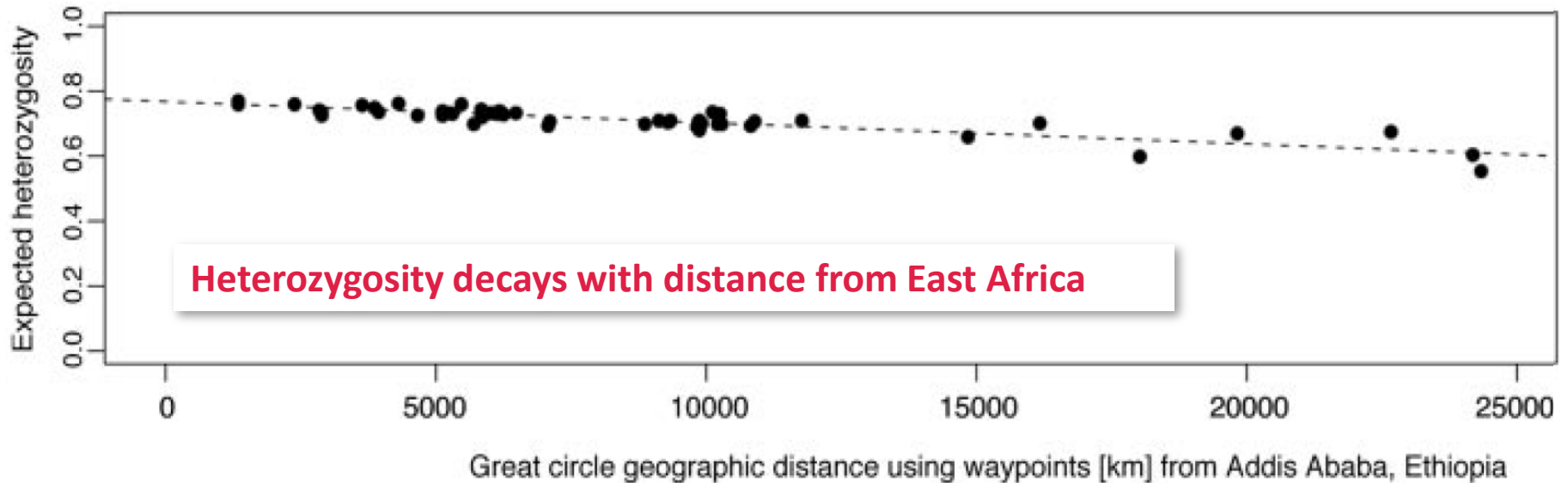
- The current genetic diversity is the outcome of past evolutionary processes.
- Hence, we can use genetic diversity to tell stories about the past.

- But this is a **challenging task!**
 - The history of natural populations is usually **complex**.
 - Several evolutionary processes can leave **similar footprints** (bottleneck vs. selection).
 - Loci are not independent, but **correlated realizations** of the same process.



Qualitative inference

- Traditionally, we have relied on qualitative inference
- **Example:** out of Africa expansion via sequential founder effects in humans.



Model-based inference

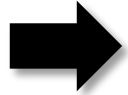
- Patterns of genetic diversity may serve as evidence for or against stories of the evolutionary past.
- Such stories are usually **vague** („Serial founder effects“).
- While the evidence may be strong, the argument remains **verbal** and is potentially **subjective**.



Model-based inference provides **statistical support**

Model-based inference

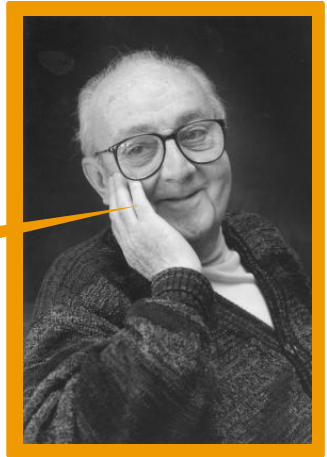
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Model-based inference provides **statistical support**

Essentially, all models are wrong, but some are useful.

George E. Box



- Qualitative inference is key when constructing sensible models!

Rejection of a Null Model

- The same as hypothesis testing in frequentist statistics:
A null model **M** is rejected using a summary statistics **s** if $\int_{s=s_{obs}}^{\infty} P(s | M) < \alpha$
- By convention, $\alpha = 0.05$
- Often the Null model is an isolated Wright-Fisher population of constant size

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Example: F-Statistics

- F_{ST} may be used to reject a panmictic population in favor of a specific structure.
- F_{IS} may be used to reject a panmictic population in favor of non-random mating (inbreeding or substructure)
- The significance of F-Statistics is usually assessed using permutation or randomization approaches.

Rejection of a Null Model: F-Statistics

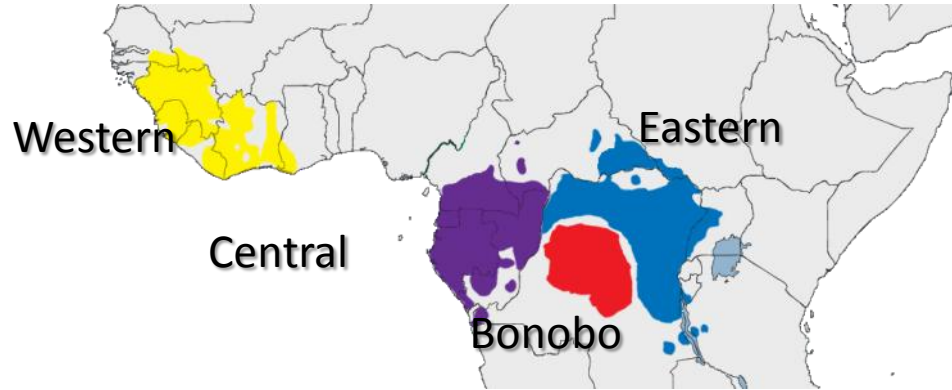
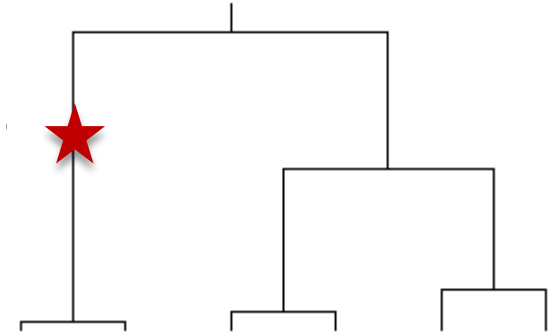


Table 1. Observed Population- and Marker-Specific F_{IS} Values.

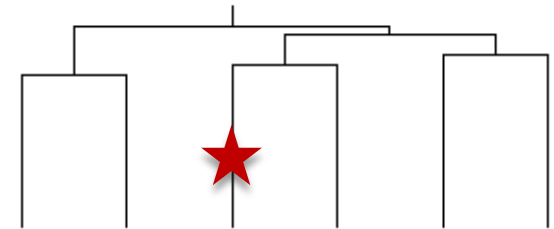
Sample	DNA	Microsatellites
Bonobo	-0.054*	0.023
Eastern chimpanzee	0.049*	0.093*
Central chimpanzee	0.111*	0.057*
Western chimpanzee	0.096*	0.026*

Rejection of a Null Model: Tajima's D

- Tajima's **D** compares two estimates of $\theta=4N\mu$ for a Wright-Fisher population of constant size:
 - one based on the number segregating sites **S**
 - one based on the average number of pairwise differences **π**
- These estimates may differ when assumptions of the Wright-Fisher population are violated.
- An expanding population, for instance, leads to a negative **D**
- Significance is usually assessed via simulations.



Wright-Fisher population



expanding population

The Felsenstein Equation

Links the observed genetic data $\mathcal{D}$ to demographic parameters Θ :

$$\mathbb{P}(\mathcal{D}|\Theta, \mu) = \int_G \mathbb{P}(\mathcal{D}|G, \mu) \mathbb{P}(G|\Theta) dG$$

Integral runs across *all possible genealogies*.

The Felsenstein Equation in Practice

- $\mathbb{P}(\mathcal{D}|G, \mu)$ is difficult to calculate.

Approximate as $\mathbb{P}(\mathcal{D}|G, \mu) \approx \mathbb{P}(S(\mathcal{D})|G, \mu)$ with summary statistics $S(\mathcal{D})$.

- The integral is impossible to solve analytically for most models.

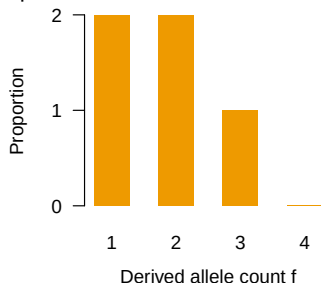
Approximate is using a random sample of genealogies:

$$P(S(\mathcal{D})|\Theta, \mu) \approx \frac{1}{N} \sum_{i=1}^N P(S(\mathcal{D})|G_i, \mu) \quad \text{where} \quad g_i \sim P(G|\Theta)$$

The Site Frequency Spectrum (SFS): our summary statistics of choice

The SFS f tabulates the sample allele frequencies of all mutations.

We have seen SFS for a single population:

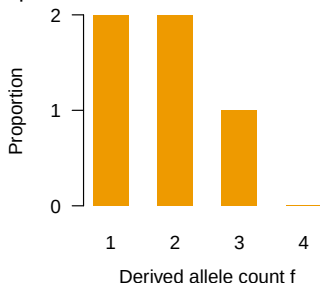


Each entry $f_1, f_2, \dots, f_{n-1}$ is the number (or proportion) of mutations at which 1, 2, ..., $n - 1$ **derived** alleles have been observed in a sample of size n .

The Site Frequency Spectrum (SFS): our summary statistics of choice

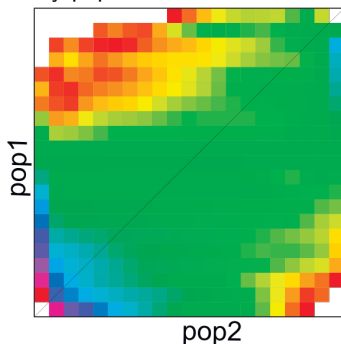
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Each entry $f_1, f_2, \dots, f_{n-1}$ is the number (or proportion) of mutations at which 1, 2, ..., $n - 1$ **derived** alleles have been observed in a sample of size n .

But they exist for arbitrary many populations:



For 2 populations, $f_{i,j}$ denotes the number of sites at i **derived** allele were observed in population 1 and j in population 2.

The Felsenstein Equation in practice

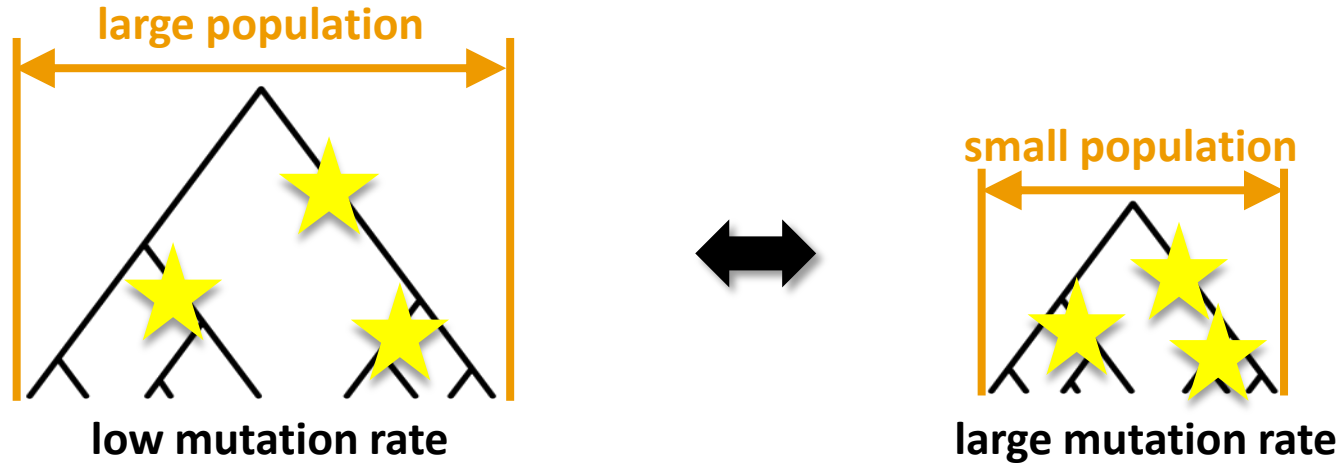
$$\mathbb{P}(\text{SFS}(\mathcal{D})|\Theta, \mu) \approx \frac{1}{N} \sum_{i=1}^N \mathbb{P}(\text{SFS}(\mathcal{D})|G_i, \mu) \quad \text{where} \quad g_i \sim P(G|\Theta)$$

Demographic inference in a nutshell

- 1) Determine the observed SFS
- 2) For a parameter vector Θ , simulate many genealogies $g_i \sim P(G|\Theta)$.
- 3) Approximate the likelihood $\mathbb{P}(\text{SFS}(\mathcal{D})|\Theta, \mu) \approx \frac{1}{N} \sum_{i=1}^N \mathbb{P}(\text{SFS}(\mathcal{D})|G_i, \mu)$
- 4) Find the parameter vector $\hat{\Theta} = \arg \max_{\Theta} \mathbb{P}(\text{SFS}(\mathcal{D})|\Theta, \mu)$ that maximizes the likelihood.

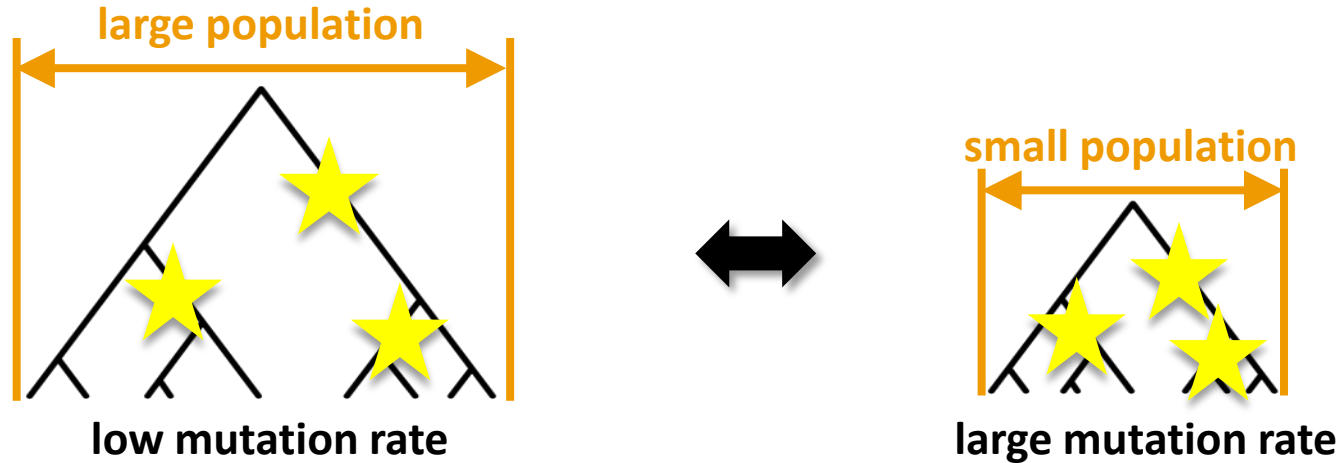
Joint inference of demography and mutation rates

- Mutation rate μ and population size N have **similar effects** on genetic diversity.



Joint inference of demography and mutation rates

- Mutation rate μ and population size N have **similar effects** on genetic diversity.



- If **sample size > effective population size**:
 - the effect of the population size is affecting the number of singletons only
 - which renders estimation of μ and N individually possible.

Deep resequencing data set

Data set:

- 202 known or prospective drug target genes
- 14,002 individuals, of which 12,514 Europeans
- Median coverage of 27x and a call rate of 90.7%



John Novembre



Matt Nelson

Extensive quality control

- Heterozygous concordance
 - 99.1% in 130 sample duplicates
 - 99.0% in comparison to 1000G Trios
- Singleton concordance
 - 98.5% in 130 sample duplicates
 - 98.3% of 245 validated via Sanger

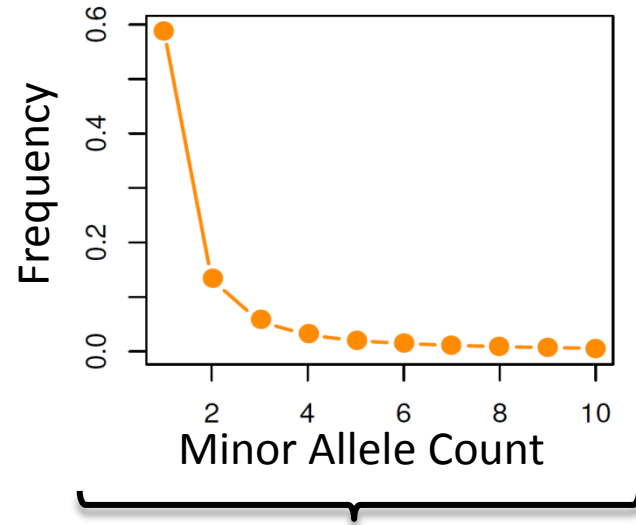
Joint inference of demography and mutation rates

- 1) Using **Site Frequency Spectrum SFS** instead of the full data **D**

AGATTCAC
AGCTTCAT
AGATTCAT
AGATTCAT
AGCTTCGC
⋮
⋮
⋮



22,000 Sequences of 202 genes

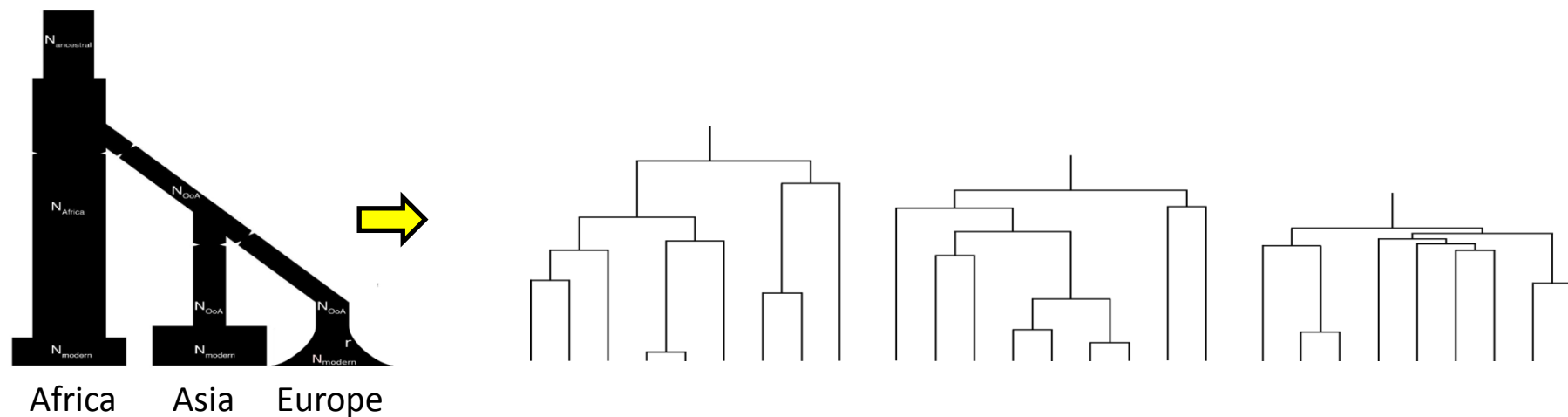


Site Frequency Spectrum SFS

Joint inference of demography and mutation rates

Using **Monte Carlo simulations** to approximate $P(\text{SFS} \mid \mu, N)$:

- Simulate genealogies with fixed parameter values

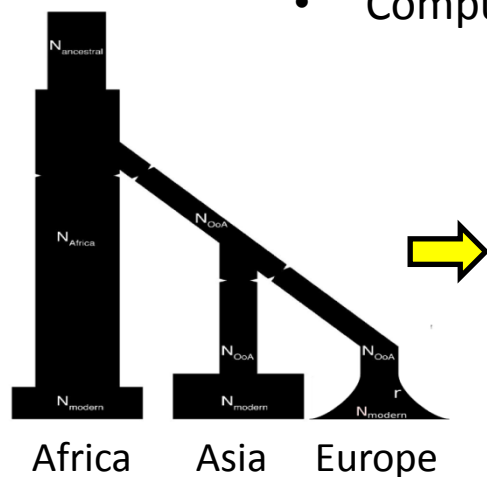


- Exponential growth in Europe
- All other parameters fixed to Schaffner estimates

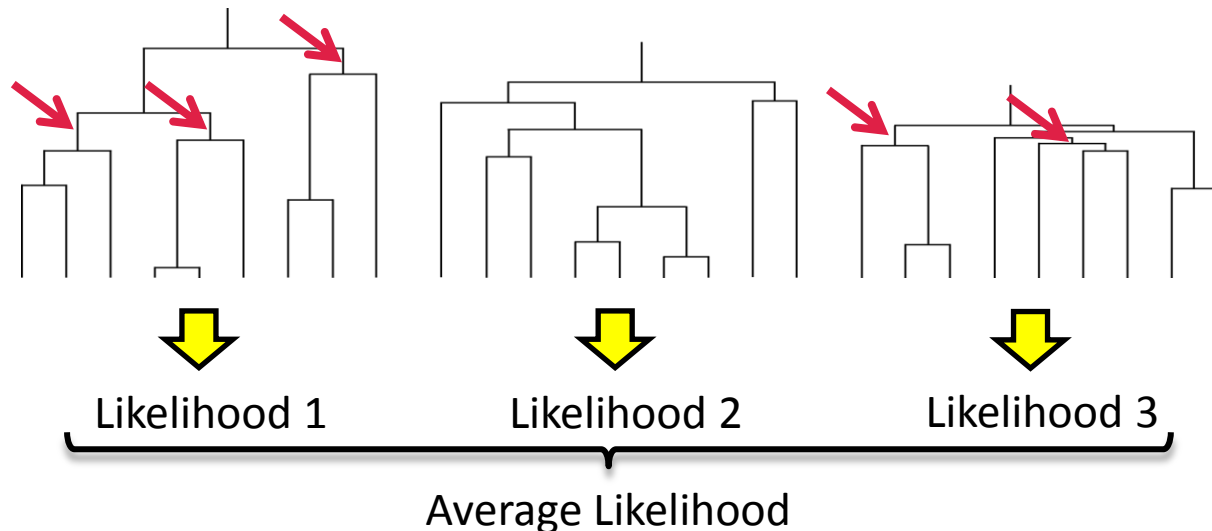
Joint inference of demography and mutation rates

Using **Monte Carlo simulations** to approximate $P(\text{SFS} \mid \mu, N)$:

- Simulate genealogies with fixed parameter values
- Compute average likelihood of the SFS across genealogies

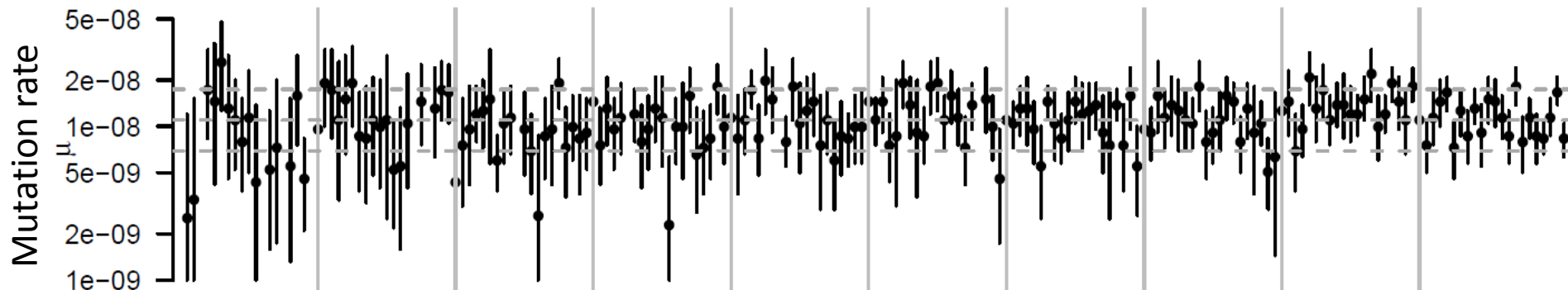
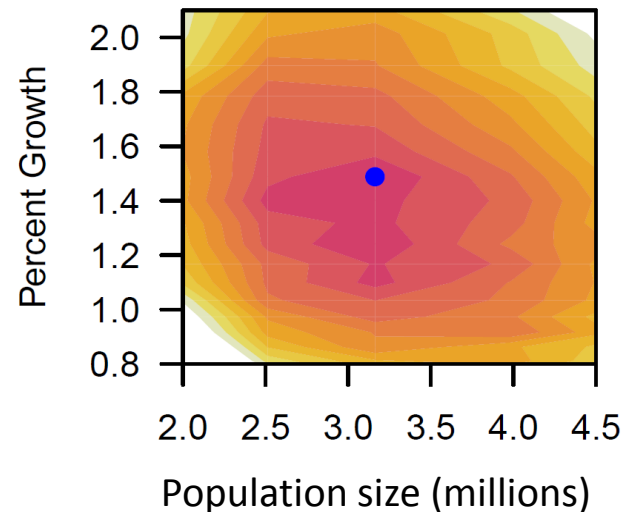


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Joint inference of demography and mutation rates

- Rapid population growth in Europe
- Variable mutation rates across genes ($p < 10^{-16}$)
- Median mutation rate of 1.2×10^{-8}
 - Lower than divergence based estimates (2.5×10^{-8})
 - But in good agreement with recent estimates from pedigrees



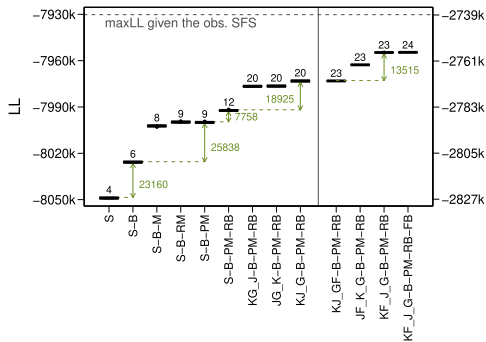
Computational efficiency of demographic inference

To render the inference efficient:

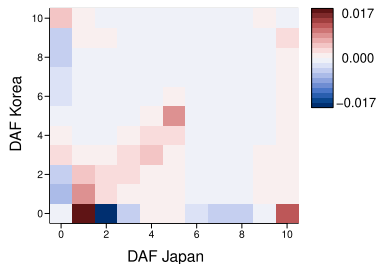
- We need an efficient way to determine the likelihood $\mathbb{P}(\text{SFS}(\mathcal{D})|\Theta, \mu)$.
- We need an efficient optimizer.

This is what is implemented in `fastsimcoal2`.

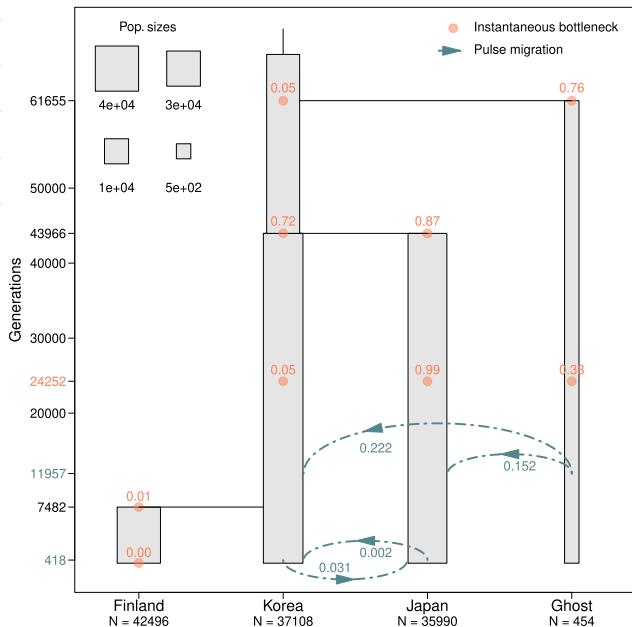
Comparison to observed LL



Δ SFS of KJ_G-B-PM-RB



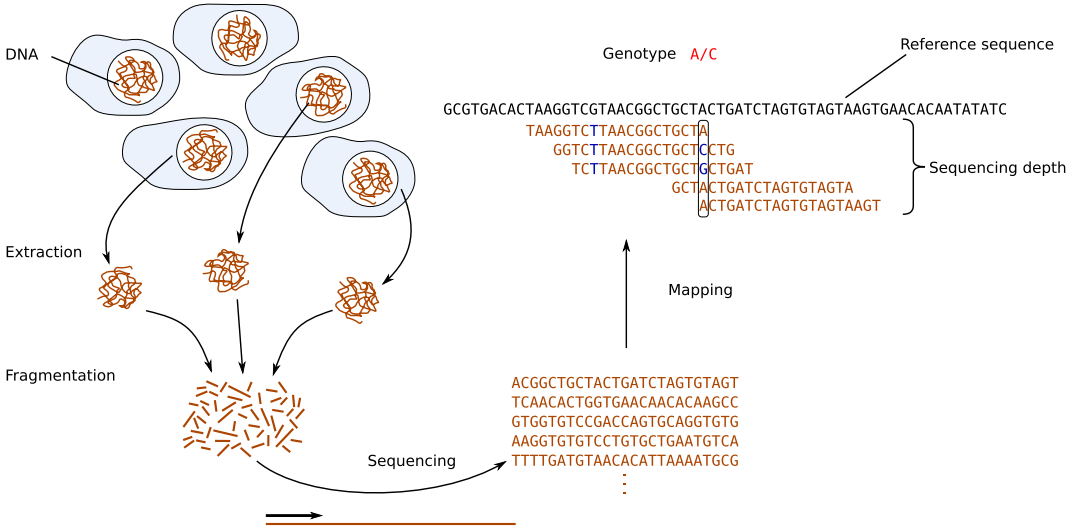
KF_J_G-B-PM-RB model plot with maxL parameter values



Remaining topic of today

How to obtain a reliable SFS from NGS data.

Standard sequencing protocol



Challenges of low-depth & ancient data

Low depth

Most positions in the genome are covered by only few reads.

```
AGGCGTGCTGACACCGTGAACACTGACACTGAACCGT
AGGCGTGCTGACACCGTGAACACTGAC
ACACCGTGAACACTGACACTGAACCGT
```

Challenges of low-depth & ancient data

Low depth

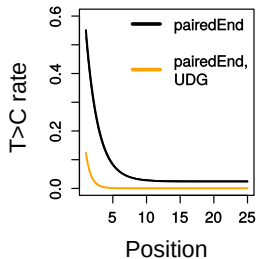
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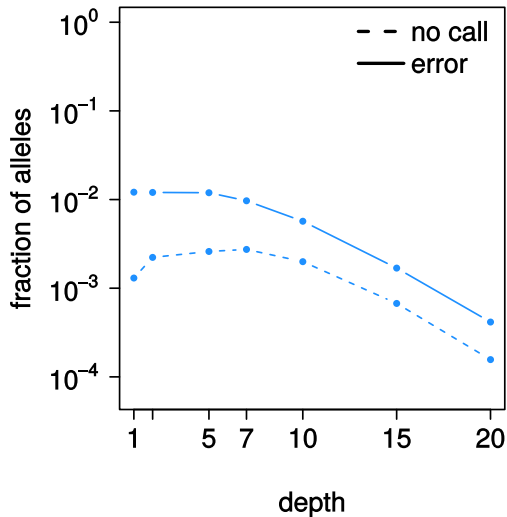
Post-Mortem Damage (PMD)

Over time, cytosins deaminate, resulting in
 $C \rightarrow T$ (or $G \rightarrow A$) errors after PCR.

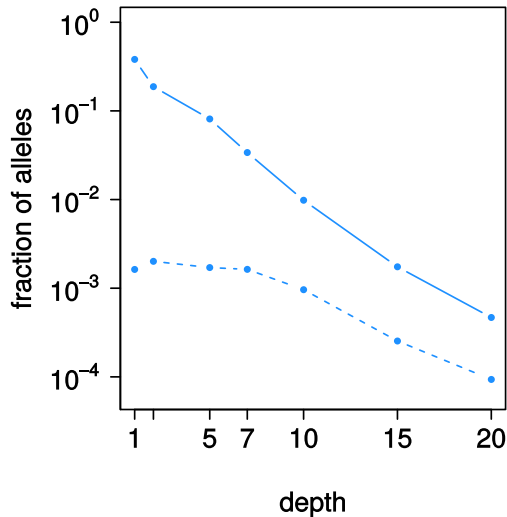


Genotyping accuracy (ML)

Homozygous



Heterozygous



Who cares about genotypes anyway?

We are rarely interested in the genotypes.

- Genetic diversity (e.g. heterozygosity).
- Association with phenotypes.
- Selection coefficients.
- ...

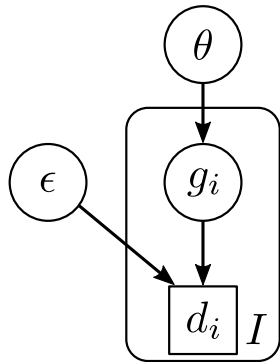
Often, genotypes are just a nuisance.

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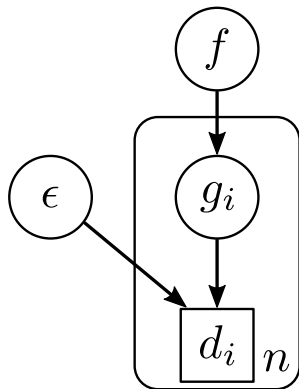
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$$\mathbb{P}(\mathbf{d}|\boldsymbol{\theta}, \epsilon) = \prod_i \sum_g \mathbb{P}(d_i|g, \epsilon) \mathbb{P}(g|\boldsymbol{\theta})$$

Example: Inferring allele frequencies



Inferring allele frequencies f under HWE

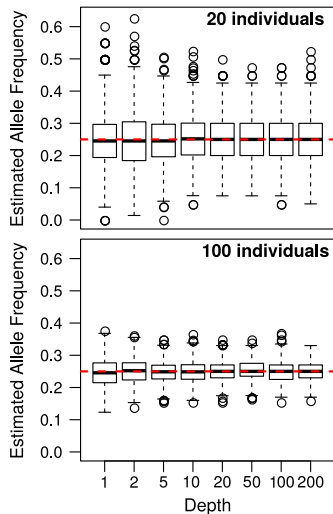
$$\mathbb{P}(\mathbf{d}|f) = \prod_n \sum_g \mathbb{P}(d_i|g, \epsilon) \mathbb{P}(g|f),$$

with

$$\mathbb{P}(g|f) = \begin{cases} (1-f)^2 & \text{if } g = 0, \\ 2f(1-f) & \text{if } g = 1, \\ f^2 & \text{if } g = 2, \end{cases}$$

The MLE is readily found using an Expectation Maximization (EM) algorithm.

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Most of the variation is biological, not technical!

Genotype likelihoods for ancient DNA

Likelihood of individual bases

$$\mathbb{P}(d_j = T | g = CC, \epsilon_j, D_{jT}) = D_{jg}(1 - \epsilon_j) + (1 - D_{jg})\epsilon_j\rho_{C \rightarrow T}$$

d_j : base in read j

g : hidden genotype

D_{jg} : generalized PMD prob

ϵ_i : error rate

$\rho_{C \rightarrow T}$: C to T rate

Inferring PMD

PMD probabilities are readily inferred from the observed changes $C \rightarrow T$ along reads.

Estimating genetic diversity from ancient DNA

Likelihood

$$L(\theta, \boldsymbol{\pi}) = \mathbb{P}(\mathbf{d}|\theta, \boldsymbol{\pi}) = \prod_{i=1}^I \sum_g \mathbb{P}(d_i|g) \mathbb{P}(g|\theta, \boldsymbol{\pi})$$

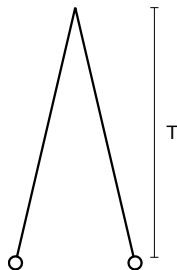
Felsenstein's substitution model (1981)

Given base frequencies $\boldsymbol{\pi} = \{\pi_A, \pi_C, \pi_G, \pi_T\}$ and the substitution rate θ ,

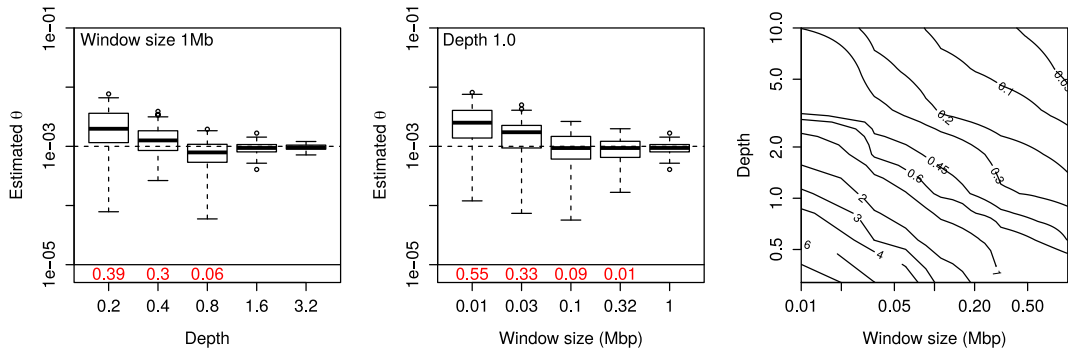
$$\mathbb{P}(g = kl|\theta, \boldsymbol{\pi}) = \begin{cases} \pi_k q_{kk}(2T) = \pi_k(e^{-\theta} + \pi_k(1 - e^{-\theta})) & \text{if } k = l, \\ \pi_k q_{kl}(2T) = \pi_k \pi_l(1 - e^{-\theta}) & \text{if } k \neq l. \end{cases}$$

Inference

We obtain ML estimate using an EM algorithm.

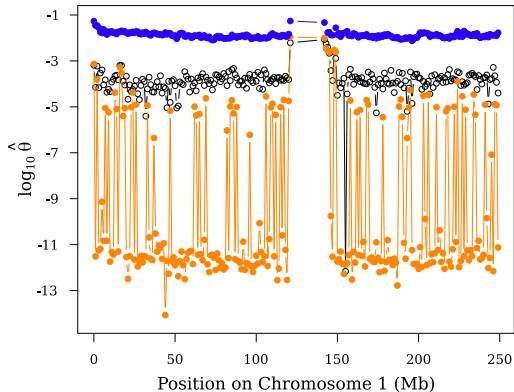


Power analysis through simulations



- Relatively high power to infer θ within a 1Mb window even at very low depth.
- Increasing window size or sequencing depth has very similar effects.

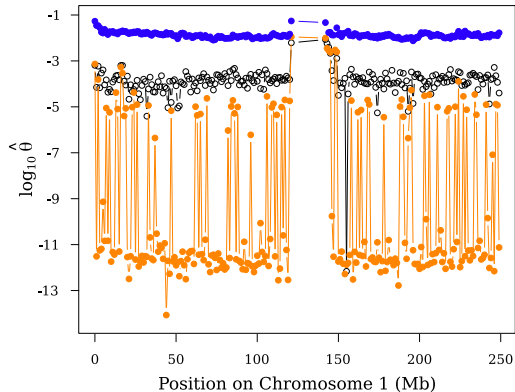
Application to ancient Greek samples



Three Greek samples

- Samples are about 10,000 years old.
- Sequencing depths from 0.8 to 3.5x.
- Expected θ for humans: $\sim 10^{-3}$.

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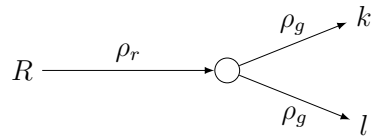
Why are these estimates so different?

We can not trust the machine provided error rates (quality scores).

Base quality recalibration

Recalibration using reference info

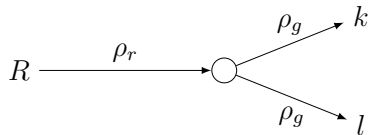
- Genotype $g = kl$ given reference R : substitution model (HKY85) with $\rho = (\rho_r, \rho_g)$.
- Reads differ due to sequencing errors (and PMD).



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Given external information $\mathbf{q}_j$ (e.g. raw quality),

$$\epsilon_j = \text{logistic} \left(\sum_k \beta_k q_{jk} \right)$$

The parameters β and π are inferred using

$$\mathbb{P}(\mathbf{d}|\beta, \pi) = \prod_l \sum_{g \in \{AA, \dots, TT\}} \mathbb{P}(g|\rho) \prod_{i=1}^n \mathbb{P}(d_{lj}|g, \epsilon_{lj}, D_{ljg})$$

Base quality recalibration

Recalibration using reference info

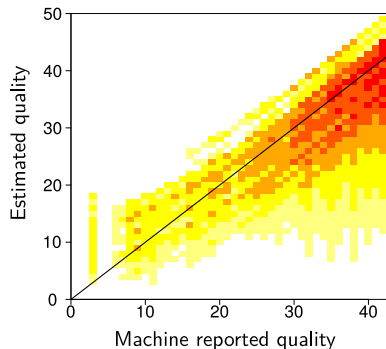
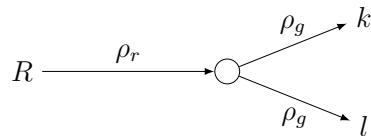
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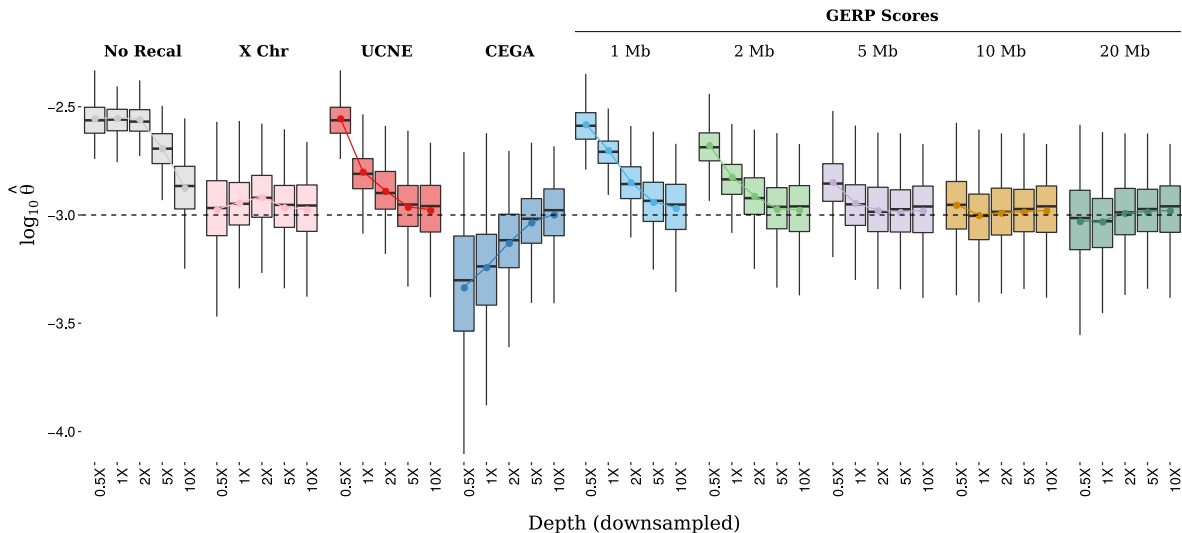
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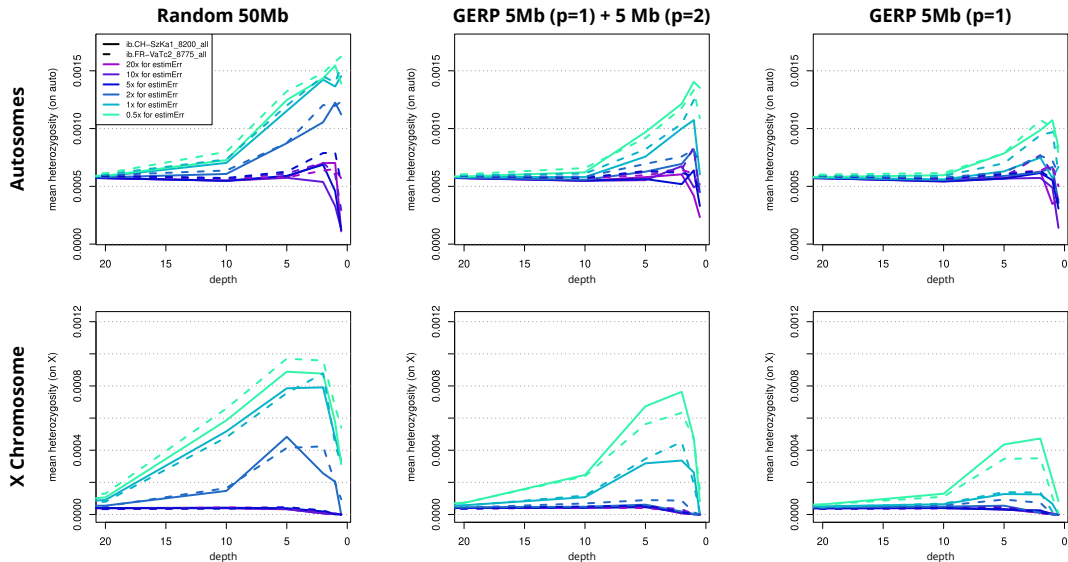
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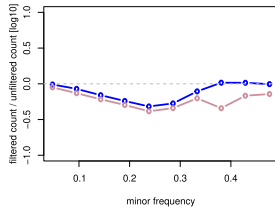
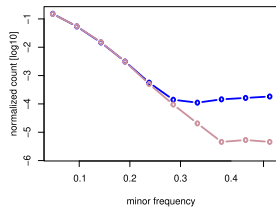
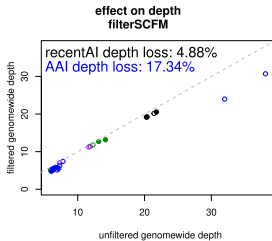
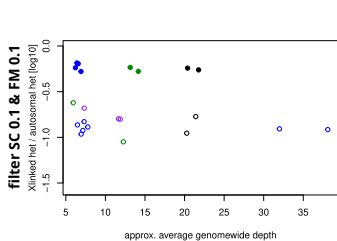
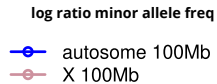
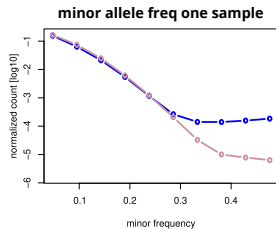
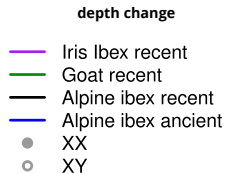
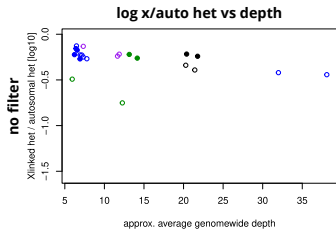
Base quality recalibration - downsampling experiments (humans)



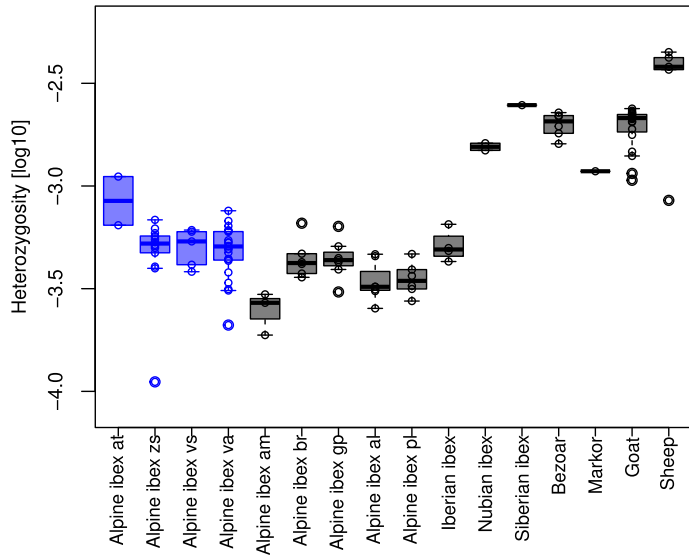
Base quality recalibration - downsampling experiments (ibex)



Relevant filters



Diversity of ancient ibex



- Very low diversity of ibex in general.
- Anthropogenic bottleneck led to further loss of diversity.

Basic ATLAS workflow:

All introduced methods are implemented in ATLAS.

We recommend the following workflow:

- 1) Merge overlapping reads.
- 2) Infer PMD and sequencing error rates.
- 3) Infer diversity statistics (e.g. θ).

Basic ATLAS workflow:

All introduced methods are implemented in ATLAS.

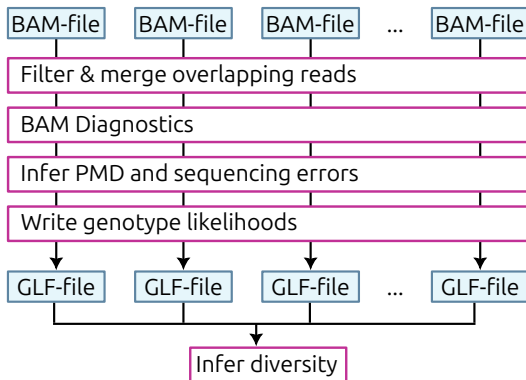
We recommend the following workflow:

- 1) Merge overlapping reads.
- 2) Infer PMD and sequencing error rates.
- 3) Infer diversity statistics (e.g. θ).

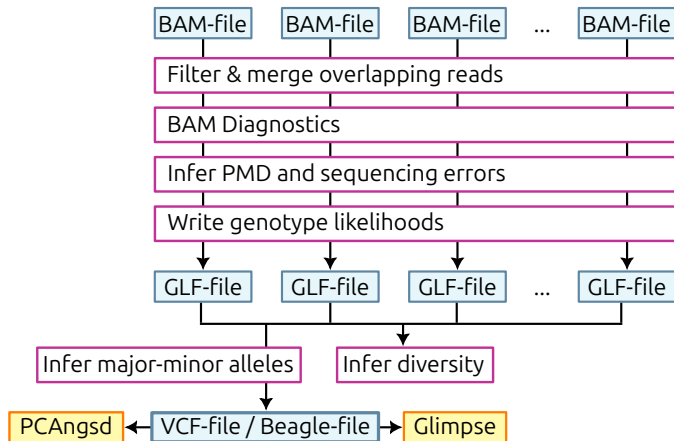
Population level analyses

However, we often have population data. How to deal with that?

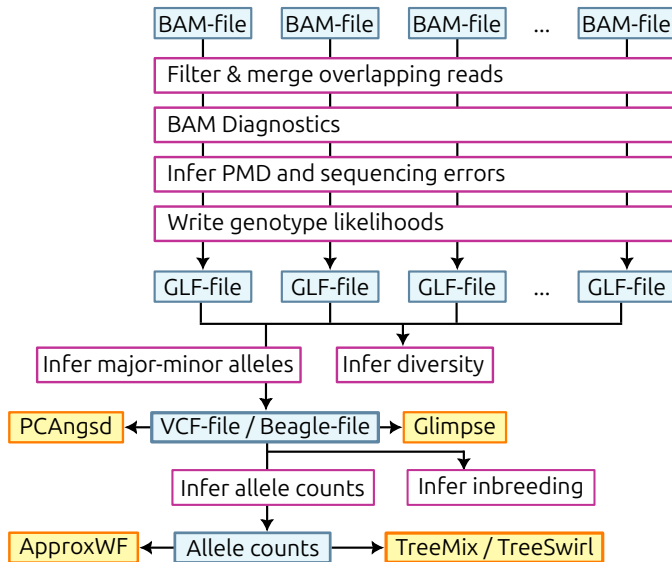
ATLAS workflow



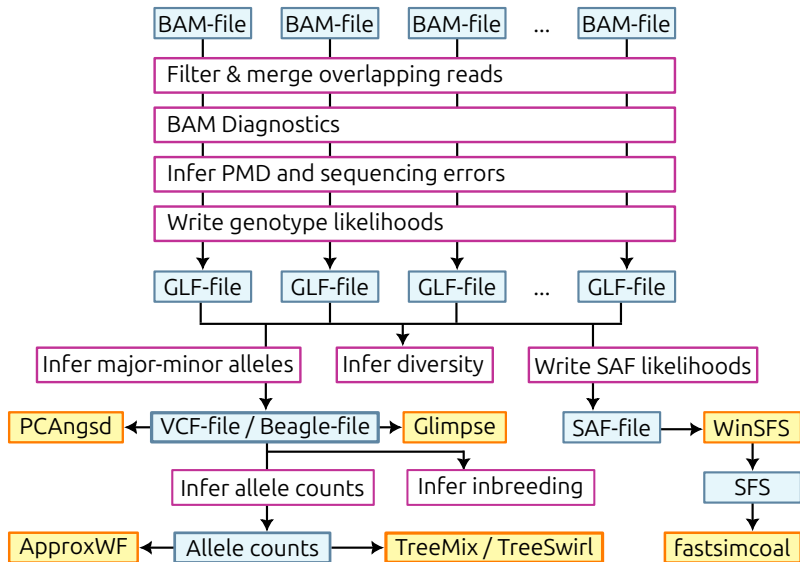
ATLAS workflow



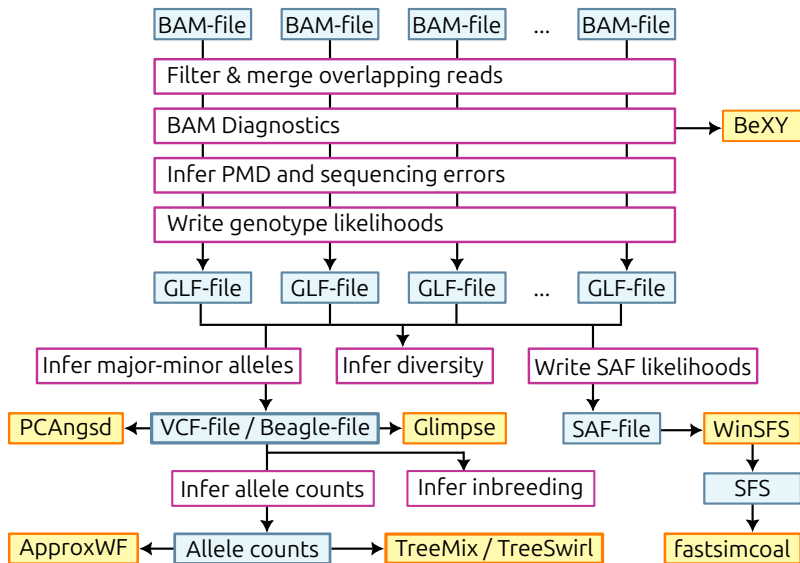
ATLAS workflow



ATLAS workflow



ATLAS workflow



ATLAS workflow → ATLAS pipeline

The ATLAS pipeline

A SnakeMake pipeline implementing all steps from fastq files to popgen analysis.

The pipeline consists of four modules:

- 1) **Gaia**: Genome Wide Alignment including Adapter-trimming.
- 2) **Rhea**: Local InDel-Realignment.
- 3) **Perses**: Post-Mortem-Damage and Error Rate Estimation for Sequence Data.
- 4) **Pallas**: Population ALLele-frequency Analysis.

Conclusions & Acknowledgments

- NGS data is noisy (and so is yours).
- Explicitly accounting for uncertainty allows for accurate inference.
- In genomics, most variation is biological, not technical. Invest in samples, not sequencing!

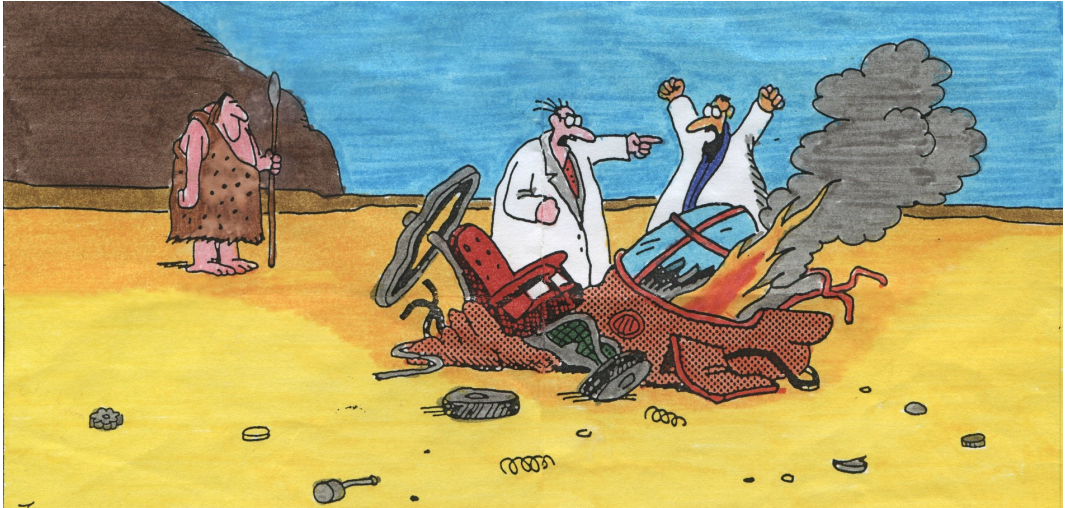
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Check for ATLAS and our other tools at <https://bitbucket.org/wegmannlab/>.

For today: check the ATLAS wiki at <https://atlaswiki.netlify.app/>.

Common Challenges in (aDNA) Science



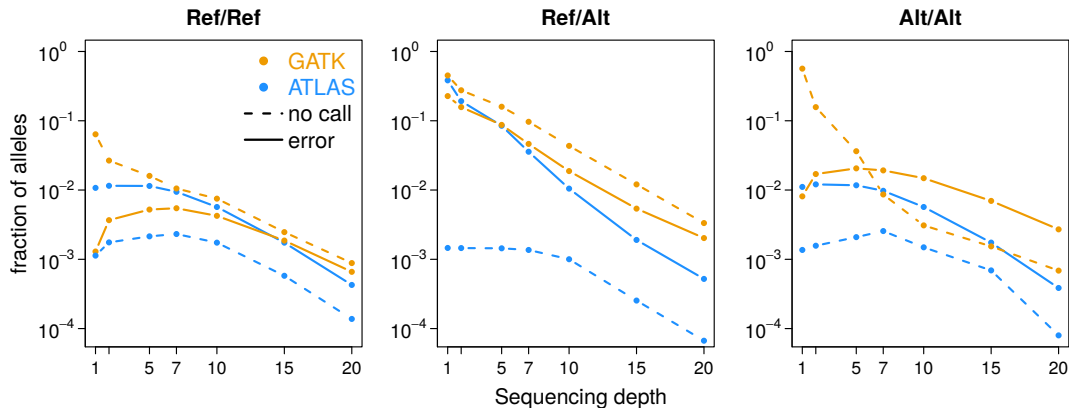
Tempers flare when Professors Carlson and Lazzall, working independently, ironically set their time machines to identical coordinates.

©Gary Larson

Genotype caller for ancient DNA

Calling ancient genotypes

- The model lends itself readily for unbiased genotype calling.
- In simulations, our caller outperforms GATK.



The Bronze Age Battle at Tollense

A massive battle

- Massive battle in approx 1250 BC involving 2,000-6,000 combatants.
- Oldest archaeological site of a battlefield of this size in Europe.



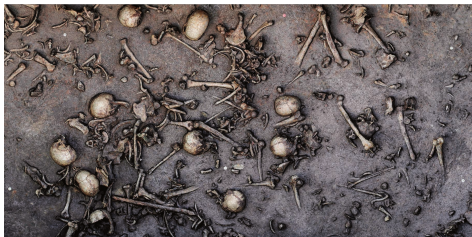
The Bronze Age Battle at Tollense

A massive battle

- Massive battle in approx 1250 BC involving 2,000-6,000 combatants.
- Oldest archaeological site of a battlefield of this size in Europe.

Genomic data for 18 soldiers

- Sequenced 5,000 unlinked, putatively neutral regions of 1000bp each (5 Mb)
- Genotyped 438 phenotypically informative markers



Were there two genetically distinct groups fighting?

Quantifying population structure

Population structure (Wahlund effect) can be quantified by the inbreeding coefficient F

$$\mathbb{P}(\mathbf{g}_i|F, p) = \begin{cases} (1 - F)(1 - p)^2 + F(1 - p) & \text{if } g_i = 0 \\ (1 - F)2p(1 - p) & \text{if } g_i = 1 \\ (1 - F)p^2 + Fp & \text{if } g_i = 2 \end{cases}$$

We estimate F and all allele frequencies p using

$$\mathbb{P}(\mathbf{d}|F, p) = \prod_l \prod_i \sum_g \mathbb{P}(\mathbf{d}_{il}|g, \epsilon_{il}, \mathbf{D}_{il}) \mathbb{P}(g|F, p_l)$$

Were there two genetically distinct groups fighting?

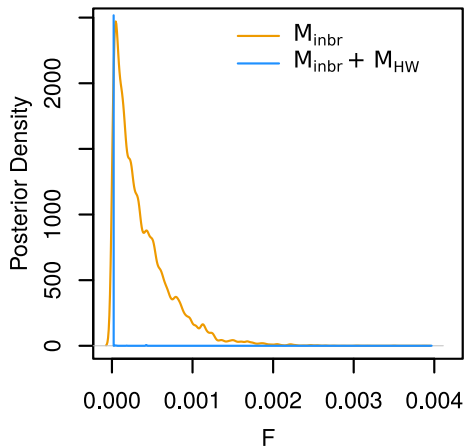
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Our samples = one population

Inferring selection at phenotypically informative markers

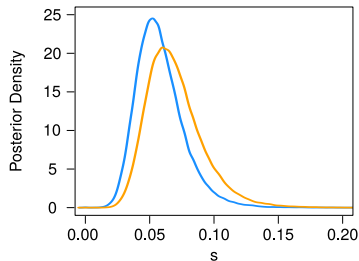
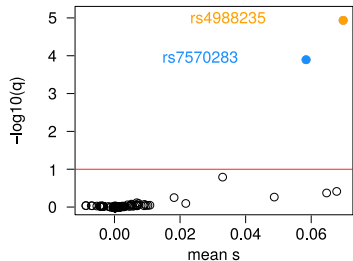
Selection at 438 markers with known association

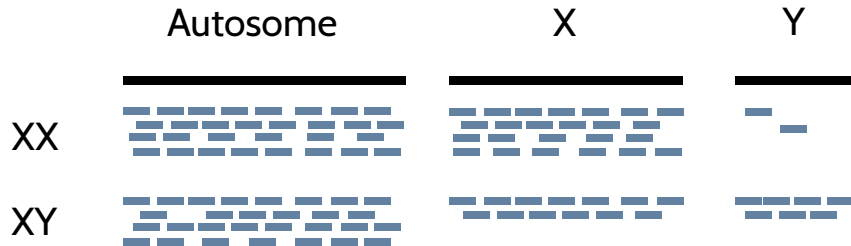
- visible phenotypes (e.g. pigmentation, morphology)
- metabolism (e.g. lactose and alcohol digestion)
- medical conditions (e.g. diabetes risk)

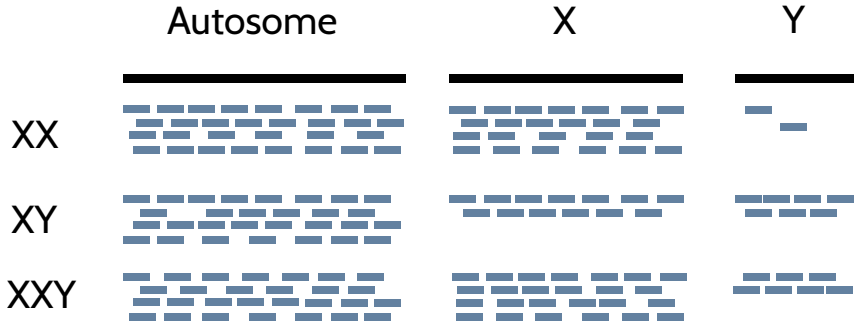
1000 Genomes CEU served as modern reference.

Strong selection on Lactase expression

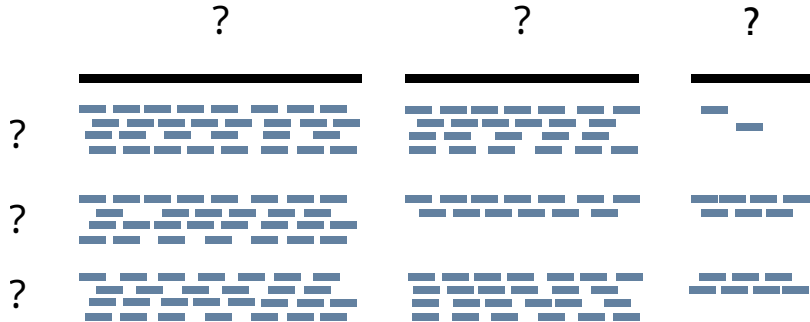
- Two markers showed strong evidence for selection, both associated with Lactase expression.
- Frequency of rs4988235: $2/32 \rightarrow 144/192$.



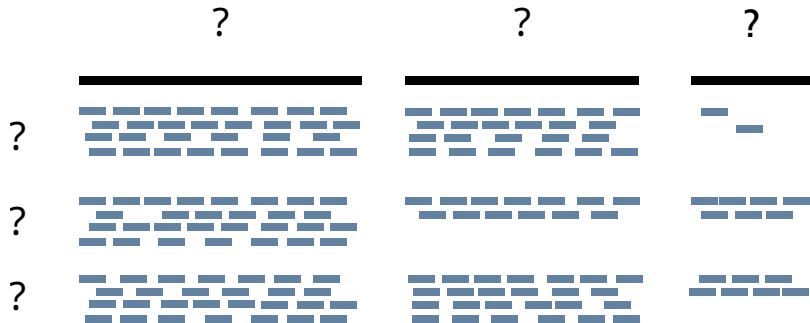




Sex karyotypes: XX, XY, X0, XXY, XYY, XXX, XXYY

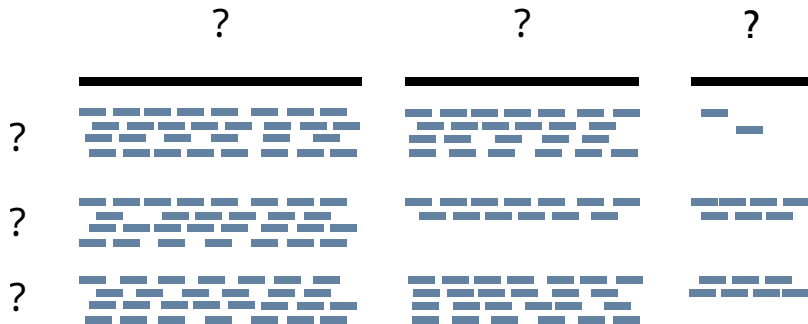


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BeXY: **B**ayesian estimation of **X** and **Y**



Sex karyotypes: XX, XY, X0, XXY, XYY, XXX, XXYY

BeXY: **B**ayesian estimation of **X** and **Y**

	c_1	c_2	c_3
i_1	30	24	2
i_2	29	13	7
i_3	28	25	7

n_{ic} : number of sequencing reads of individual i mapped to scaffold c

We model the vector of read counts $\mathbf{n}_i = (n_{i1}, \dots, n_{iC})$ as

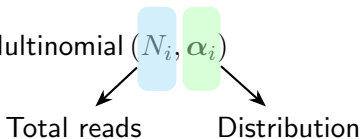
$$\mathbf{n}_i \sim \text{Dirichlet-Multinomial}(N_i, \boldsymbol{\alpha}_i)$$

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Total reads

We model the vector of read counts $\mathbf{n}_i = (n_{i1}, \dots, n_{iC})$ as

$$\mathbf{n}_i \sim \text{Dirichlet-Multinomial} (N_i, \boldsymbol{\alpha}_i)$$


The diagram illustrates the components of the Dirichlet-Multinomial distribution. The parameter N_i is highlighted with a light blue rounded rectangle, and an arrow points from it to the text "Total reads". The parameter $\boldsymbol{\alpha}_i$ is highlighted with a light green rounded rectangle, and an arrow points from it to the text "Distribution".

We model the vector of read counts $\mathbf{n}_i = (n_{i1}, \dots, n_{iC})$ as

$$\mathbf{n}_i \sim \text{Dirichlet-Multinomial}(\mathbf{N}_i, \boldsymbol{\alpha}_i)$$

Total reads Distribution

$$\alpha_{ic} = \frac{1}{\sigma} \gamma_c (p_{ic} + \epsilon_i),$$

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$$\mathbf{n}_i \sim \text{Dirichlet-Multinomial}(\underbrace{N_i}_{\text{Total reads}}, \underbrace{\boldsymbol{\alpha}_i}_{\text{Distribution}})$$

$$\alpha_{ic} = \frac{1}{\sigma} \underbrace{\gamma_c}_{\text{Mapping attractor}} (p_{ic} + \epsilon_i),$$

BeXY - Model

We model the vector of read counts $\mathbf{n}_i = (n_{i1}, \dots, n_{iC})$ as

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Mapping attractor Ploidy

We model the vector of read counts $\mathbf{n}_i = (n_{i1}, \dots, n_{iC})$ as

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Mapping attractor Ploidy Noise

We model the vector of read counts $\mathbf{n}_i = (n_{i1}, \dots, n_{iC})$ as

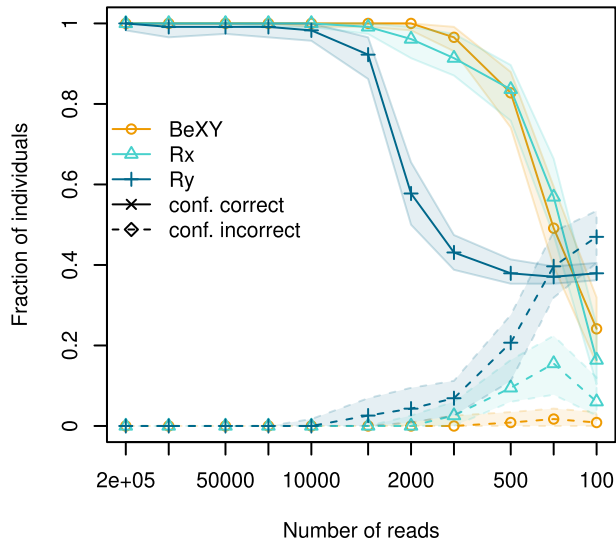
$$\mathbf{n}_i \sim \text{Dirichlet-Multinomial}(N_i, \boldsymbol{\alpha}_i)$$

Total reads Distribution

$$\alpha_{ic} = \frac{1}{\sigma} \gamma_c (p_{ic} + \epsilon_i),$$

Scale Mapping attractor Ploidy Noise

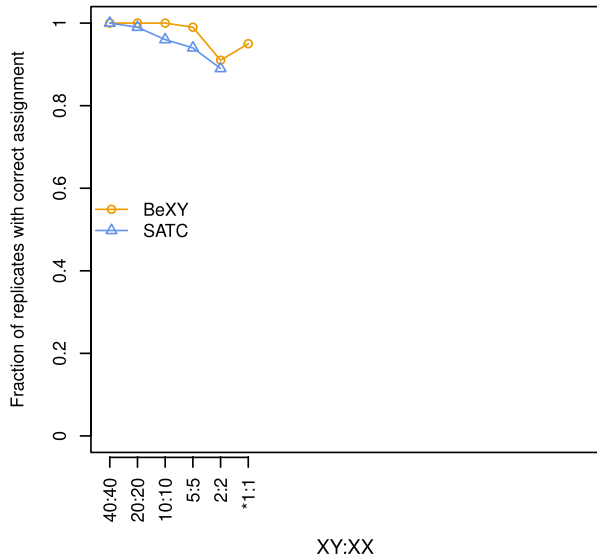
BeXY - Sexing



R_y : Skoglund *et al.* (2013)

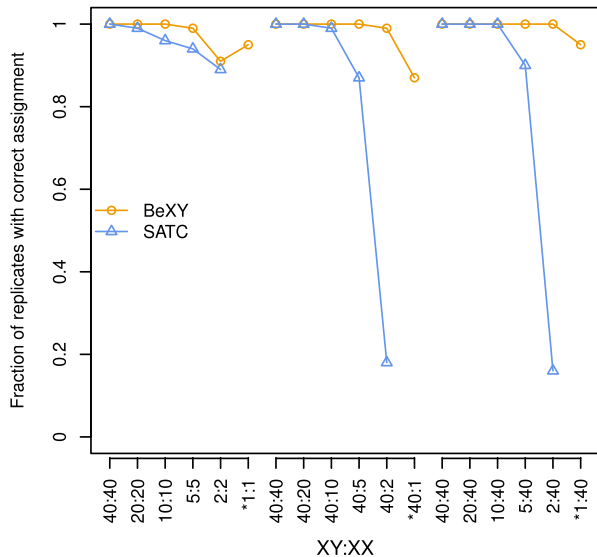
R_x : Mitnik *et al.* (2016)

BeXY - Joint inference



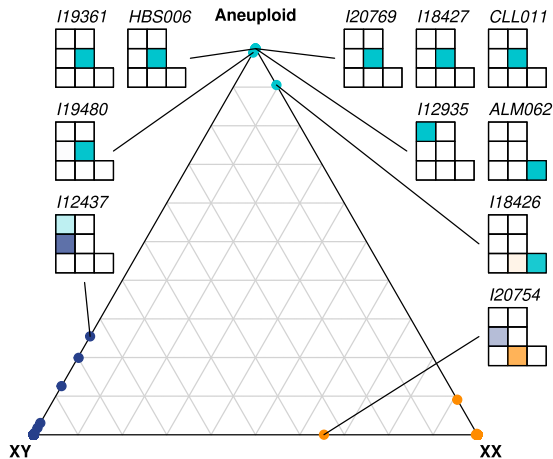
SATC: Nursyifa *et al.* (2022)

BeXY - Joint inference



SATC: Nursyifa *et al.* (2022)

BeXY - Ancient humans ($n = 2,457$)



YY	XYX	XXYY	
Y	XY	XXY	
0	X0	XX	XXX
	X	XX	XXX

- 2,457 ancient humans
- 1240k Capture data
- Aneuploid: 0.37%
- Figure generated by R-package